SMART TRAFFIC VISION

AI-Based Traffic Monitoring and Violation Detection System

1. Project Overview

Smart Traffic Vision is a Computer Vision and Artificial Intelligence-based system designed to automatically monitor traffic using **images, recorded videos, and potentially live CCTV/camera feeds**. The system will detect and classify vehicles, track their movement, count the number of vehicles passing through a traffic area, and identify selected traffic-rule violations.

The proposed system will use **Python, OpenCV, YOLO, object tracking algorithms, OCR, and Streamlit**. It will provide a dashboard showing vehicle counts, traffic statistics, detected violations, and captured evidence.

The project can also be extended with a **number-plate recognition and simulated challan module**, making it useful as an academic prototype for intelligent traffic monitoring.

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2. Problem Statement

Traditional traffic monitoring depends heavily on manual observation of CCTV cameras. Monitoring a large number of vehicles continuously can be difficult and may result in missed violations.

The proposed system aims to automate traffic analysis by using Computer Vision to detect vehicles and identify selected violations such as:

- No helmet for two-wheelers
- No seat belt where camera conditions permit
- Red-light violation
- Wrong-way driving

The system can also count vehicles and maintain evidence of detected violations.

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3. Proposed System

The system will follow the pipeline:

Traffic Image / Video / Camera → Vehicle Detection → Vehicle Classification → Vehicle Tracking → Vehicle Counting → Violation Detection → Number Plate OCR → Evidence Generation → Dashboard

Main Features

- 🚙 Automatic vehicle detection and classification
- 🔢 Vehicle counting using virtual line/region
- 🎥 Recorded video and image analysis
- 📡 Future support for live CCTV/webcam
- 🏍️ Helmet detection
- 🚗 Seat-belt detection under suitable camera conditions
- 🚦 Red-light violation detection
- ↔ Wrong-way detection
- 🔍 Number-plate detection and OCR
- 📸 Automatic violation evidence capture
- 📊 Traffic and violation analytics dashboard
- 📄 Simulated digital challan/violation record

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4. Technologies and Algorithms

Component	Technology / Algorithm
Programming	Python
Image/Video Processing	OpenCV
Vehicle Detection	YOLO
Object Tracking	ByteTrack / BoT-SORT
Number Plate OCR	EasyOCR / PaddleOCR
Data Processing	NumPy, Pandas
Database	SQLite
Dashboard	Streamlit

Basic Working

YOLO will detect vehicles in each video frame.
ByteTrack/BoT-SORT will assign a unique ID to each vehicle and track it across multiple frames. This will allow us to count each vehicle only once when it crosses a predefined virtual line.
For violations, separate Computer Vision modules will analyze the detected vehicle/rider. For example, a two-wheeler can be checked for helmet usage, while vehicle movement can be analyzed for red-light and wrong-way violations.
For number-plate recognition, the detected plate will be cropped and processed using OCR to extract the registration number.

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5. Expected Output

For example, while processing a traffic video, the system may display:
Total Vehicles: 250
Cars: 130
Two-Wheelers: 95
Buses: 10
Trucks: 15
Detected Violations:

- No Helmet: 18
- No Seat Belt: 5
- Red-Light Violation: 7
- Wrong-Way: 3

For every detected violation, the system can save an evidence frame containing the vehicle, violation type, timestamp, and vehicle number if successfully detected.

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6. Future Scope

The project can later be extended to:

- Real-time CCTV/IP camera monitoring
- Multi-camera traffic monitoring
- Advanced Automatic Number Plate Recognition (ANPR)
- Traffic congestion and density analysis
- Vehicle speed estimation
- Lane violation detection
- Illegal parking detection
- Mobile/web application

- Cloud-based monitoring
- Traffic prediction and analytics
- Integration with authorized traffic-management systems

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7. Conclusion

Smart Traffic Vision aims to demonstrate how Artificial Intelligence and Computer Vision can be used for automated traffic monitoring. Instead of manually monitoring every vehicle, the proposed system can automatically detect, classify, track, and count vehicles while identifying selected traffic violations and generating visual evidence.

The project will initially focus on **vehicle detection, tracking, counting, and selected violation detection using recorded traffic videos**, and will gradually be extended toward real-time camera processing and number-plate-based violation records.